Vundyala Sreekar Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.71
Sri Chaitanya Junior College	Class XII (Board of Intermediate Education Telangana)	2023	96.50%
Keshavareddy concept school	Class X (Board of Secondary Education Telangana)	2021	10.00%

PROJECTS

Real-Time Multiplayer Poker Platform | Python, FastAPI, WebSockets, SQLite, Docker, pytest | LINK

- Built a **server-authoritative** Texas Hold'em web app with private rooms, full betting logic (blinds, side pots, all-ins) and real-time multiplayer play over **WebSockets**, with hidden cards never exposed to other clients
- Optimized 7-card hand evaluation from **~11,800 to ~214,000 hands/sec (~18x)** by replacing a combinations-based approach with **bitmask-based rank/suit encoding**; verified identical to the original across 100,000 random hands
- Diagnosed and fixed a production-blocking **asyncio deadlock** in the WebSocket test suite (event-loop/thread contention); added pytest-timeout as a permanent safeguard
- Achieved **98% test coverage** across **139 tests**, including a fuzz-testing suite simulating thousands of randomized hands with **zero chip-conservation violations**; deployed via **Docker** on Render with **CI** on every push

AI-Powered Research Paper Retrieval Engine | C++, Python, FastAPI, cpp-httplib, Docker, Cohere API

- Built a **custom C++ search engine** (trie, inverted index, TF-IDF) served over REST via cpp-httplib
- Added a **FastAPI** microservice for Cohere embeddings and generation, coordinating with the C++ engine
- Fused TF-IDF and Cohere embedding rankings via **reciprocal rank fusion** over live Semantic Scholar results
- Built a **RAG pipeline** with Cohere giving cited answers that **decline to answer** when no relevant excerpts are found
- Deployed **two Dockerized services** on Render with a live frontend for search, hybrid ranking, and Q&A

Mini Ride-Hailing Backend | C++, STL, OOP

- Built a backend simulation of a ride-booking system with driver matching, ride lifecycle, and fare calculation
- Implemented nearest-driver allocation using **Min-Heap (Priority Queue)** on Euclidean distance
- Used unordered_map for O(1) user/driver lookups and dynamic driver location updates after each trip
- Tested edge cases including unregistered users, unavailable drivers, and multi-user ride history queries
- Designed modular class architecture across User, Driver, Ride, and RideManager with clean separation of concerns

Braking System Design and Manufacturing - Formula Student Car (Aeolus Racing)| SolidWorks, FEA, CNC

- Engineered the braking system layout ensuring compliance with Formula SAE rules and optimal brake bias
- Selected OEM calipers and master cylinders based on required brake torque, pedal ratio, and packaging constraints
- Designed a ventilated brake disc in SolidWorks, optimizing heat dissipation, weight, and manufacturability
- Led CNC manufacturing of the disc; validated with FEA, brake torque calculations, and on-car testing

SKILLS

Languages: C++, Python, SQL

Core CS: DSA, OOP, Operating Systems, DBMS, Computer Networks

Backend & AI: REST APIs, FastAPI, Microservices, RAG, Embeddings, Semantic Search

Tools: Git/GitHub, Docker, Linux, Render

Engineering: SolidWorks, FEA,CFD, CNC Manufacturing

POSITIONS OF RESPONSIBILITY

Sponsorship Head - Milan, General Championship of IIT Hyderabad

- Raised **3.5L+** in cash and in-kind sponsorship from 6 partners, including Skechers, ICICI and SafExpress
- Led a 10-member team running outreach to ~500 companies; personally pitched and negotiated with ~20 prospects

Brakes Domain Head - Vehicle Dynamics, Aeolus Racing (Racing Club, IIT Hyderabad)

- Led the brakes domain through design, fabrication and assembly of the Formula Student car's braking system
- Taught 3 juniors load transfer, disc design and analysis; reviewed and verified their work

UG Mentor - Sunshine, Counselling Cell, IIT Hyderabad

- Mentored ~20 first-year students on academic planning, course selection and campus resources

CERTIFICATIONS

Google Cybersecurity Professional Certificate -- Coursera (Google), Aug 2024

- Linux, SQL, Python, SIEM, IDS

Formula Student Car Design & Development -- 6-week internship program, **FMAE Academy**, Nov-Dec 2024

EXTRACURRICULAR

Corporal rank holder in **NCC**